

JUNGFRAUJOCH, Switzerland

WMO: 067300

Lat: 46.5475N

Lon: 7.9853E

Elev: 3576

StdP: 65.12

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-24.8	-23.0	-36.9	0.2	-10.5	-34.2	0.2	-10.4	25.3	-14.5	22.6	-13.6	7.4	310	1.156

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.4	7.3	2.0	6.1	1.4	5.0	0.9	3.8	5.8	2.9	4.7	2.1	3.8	3.4	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2.7	7.2	4.1	1.9	6.7	3.3	1.1	6.4	2.7	23.4	5.9	21.2	4.8	19.4	3.9	7.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.0	18.8	15.7	DB	-26.5	10.4	2.3	1.6	-28.1	11.5	-29.4	12.5	-30.7	13.4	-32.3	14.6	(4)
(5)			WB	-26.8	5.7	2.1	1.1	-28.3	6.6	-29.5	7.2	-30.7	7.8	-32.2	8.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-6.5	-12.8	-13.2	-11.4	-9.1	-4.9	-1.1	0.6	0.9	-2.3	-4.4	-9.3	-11.6
	DBStd	6.52	4.43	4.99	4.49	3.92	3.52	3.71	3.28	3.17	3.76	4.03	4.76	4.72
	HDD10.0	6023	707	651	663	574	461	332	291	281	369	447	578	669
	HDD18.3	9064	966	884	921	824	719	582	549	539	619	705	828	928
	CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	6.8	7.9	7.7	7.6	7.1	6.7	5.9	5.5	5.1	5.6	6.8	8.0	8.2
Precipitation	PrecAvg	1327	81	84	85	97	141	145	155	156	105	87	97	96
	PrecMax	1648	200	249	185	195	241	237	320	300	173	149	251	251
	PrecMin	987	13	26	33	35	52	68	66	37	49	2	7	3
	PrecStd	142	47	53	37	35	56	40	57	52	32	36	50	52
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.6	-0.5	-1.0	0.8	4.3	8.9	8.5	9.2	6.7	4.4	1.7	-0.9
		MCWB	-6.7	-6.3	-5.4	-4.4	-0.1	2.0	3.7	2.6	-0.1	-2.2	-4.5	-5.3
	2%	DB	-4.0	-2.9	-3.1	-1.2	2.3	6.5	6.8	7.3	4.9	2.7	-0.3	-3.5
		MCWB	-8.1	-7.8	-7.4	-5.4	-1.6	1.5	2.4	1.9	-1.4	-2.8	-5.2	-7.7
	5%	DB	-5.4	-4.8	-4.3	-2.7	0.8	4.8	5.6	6.0	3.6	1.5	-1.9	-4.7
		MCWB	-8.4	-8.4	-7.9	-5.7	-2.2	0.9	1.7	1.5	-1.7	-3.3	-6.0	-8.2
	10%	DB	-6.9	-6.5	-5.7	-4.2	-0.5	3.5	4.5	4.8	2.4	0.4	-3.4	-5.8
		MCWB	-9.2	-9.1	-8.4	-6.5	-2.9	0.4	1.1	1.0	-1.9	-3.7	-6.6	-8.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-5.3	-5.2	-4.0	-2.8	1.2	4.3	5.1	5.1	2.2	0.3	-2.0	-4.3
		MCDB	-4.6	-2.6	-2.6	-1.0	3.0	6.5	7.0	7.0	3.8	1.4	-0.8	-2.8
	2%	WB	-6.8	-6.8	-5.8	-4.1	-0.4	2.9	3.8	3.8	1.2	-0.8	-3.8	-5.9
		MCDB	-5.3	-4.7	-4.5	-2.5	1.1	4.8	5.7	5.8	2.8	0.6	-2.4	-5.0
	5%	WB	-7.8	-7.9	-7.0	-5.1	-1.3	1.9	2.9	2.9	0.3	-1.7	-4.9	-7.2
		MCDB	-6.1	-5.4	-5.5	-3.6	0.0	3.8	4.7	4.7	1.7	-0.1	-3.1	-5.9
	10%	WB	-9.0	-9.0	-8.0	-6.1	-2.3	1.0	2.1	2.0	-0.6	-2.6	-5.9	-8.3
		MCDB	-7.1	-6.5	-6.3	-4.7	-1.0	2.9	3.7	3.7	0.8	-0.8	-4.0	-6.8
Mean Daily Temperature Range	5% DB	MDBR	5.3	5.2	5.1	5.1	4.9	4.9	4.6	4.4	4.4	4.5	4.6	5.1
		MCDBR	6.0	6.0	5.8	6.2	6.1	6.1	5.8	5.5	5.5	4.9	5.2	5.6
		MCWBR	4.8	4.5	4.6	5.2	5.2	5.5	5.2	5.1	4.7	3.7	4.0	4.8
	5% WB	MCDBR	6.2	6.2	5.3	5.8	5.3	5.4	5.2	4.9	4.7	4.4	4.6	5.9
		MCWBR	6.0	5.5	5.2	5.3	4.9	5.3	5.1	4.8	4.7	4.4	4.5	6.0
Clear-Sky Solar Irradiance	taub		0.180	0.194	0.220	0.246	0.259	0.265	0.265	0.259	0.244	0.227	0.201	0.179
	taud		2.446	2.417	2.412	2.459	2.519	2.539	2.528	2.564	2.607	2.635	2.579	2.469
	Ebn at Noon		989	1027	1031	1021	1009	999	995	994	992	974	953	956
	Edh at Noon		62	81	96	102	101	99	99	91	79	65	55	54
All-Sky Solar Radiation	RadAvg		1.48	2.40	3.56	4.41	4.88	5.77	5.72	4.97	3.89	2.55	1.50	1.22
	RadStd		0.18	0.31	0.51	0.67	0.46	0.53	0.49	0.45	0.36	0.28	0.21	0.17

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.30	N/A	-1.86	+0.77	+0.50	+0.38	N/A	N/A	N/A	N/A
Regional (118 neighbors)		N/A	N/A	N/A	+0.72	+0.42	+0.32	N/A	N/A	+57	+16

Nomenclature: See separate page